

Distinct Lineage Specification Roles for NANOG, OCT4, and SOX2 in Human Embryonic Stem Cells

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SUMMARY

Nanog, Oct4, and Sox2 are the core regulators of mouse (m)ESC pluripotency. Although their basic importance in human (h)ESCs has been demonstrated, the mechanistic functions are not well defined. Here, we identify general and cell-line-specific requirements for NANOG, OCT4, and SOX2 in hESCs. We show that OCT4 regulates, and interacts with, the BMP4 pathway to specify four developmental fates. High levels of OCT4 enable self-renewal in the absence of BMP4 but specify mesendoderm in the presence of BMP4. Low levels of OCT4 induce embryonic ectoderm differentiation in the absence of BMP4 but specify extraembryonic lineages in the presence of BMP4. NANOG represses embryonic ectoderm differentiation but has little effect on other lineages, whereas SOX2 and SOX3 are redundant and repress mesendoderm differentiation. Thus, instead of being panrepressors of differentiation, each factor controls specific cell fates. Our study revises the view of how self-renewal is orchestrated in hESCs.

INTRODUCTION

mESCs rely on a complex network of transcription factors to maintain the pluripotent state (Loh and Lim, 2011). Nanog, Oct4, and Sox2 are the core components of this network. The POU domain protein Oct4 (Pou5f1) specifies the inner cell mass (ICM) during embryonic development by counteracting differentiation into trophectoderm (TE) (Nichols et al., 1998). Carefully balanced Oct4 levels are required for the maintenance of the pluripotent state in vitro. Inactivation of Oct4 in mESCs causes rapid loss of self-renewal and induction of TE differentiation, whereas Oct4 overexpression triggers differentiation into endoderm (EN) and mesoderm (ME) (Ivanova et al., 2006; Niwa et al., 2000). The homeodomain-containing protein Nanog controls the epiblast versus primitive endoderm (pEN) decision in the blastocyst (Chazaud et al., 2006). Nanog-depleted blastocysts fail to establish ESC lines and differentiate into pEN in culture (Mitsui et al., 2003). Downregulation of Nanog in mESCs yields a broader repertoire of cell lineages, with primary contribu-

tion to TE and pEN (Ivanova et al., 2006). The SRY-related HMG box protein Sox2 is required for epiblast maintenance (Avilion et al., 2003) and Sox2-depleted mESCs differentiate into TE (Ivanova et al., 2006; Masui et al., 2007).

Per the prevailing model (Jaenisch and Young, 2008), Nanog, Oct4, and Sox2 cooperatively maintain the regulatory network responsible for self-renewal and pluripotency. Molecular analyses in mESCs have demonstrated that Nanog, Oct4, and Sox2 coregulate large cohorts of genes (Ivanova et al., 2006) and co-occupy many regulatory loci (Boyer et al., 2005; Loh et al., 2006). Cobinding of these factors at the active enhancers recruits the global transcriptional coactivator p300 (Chen et al., 2008), whereas cobinding at the promoters of developmental regulators mediates gene silencing (Boyer et al., 2005).

Given the key roles of Nanog, Oct4, and Sox2 in mESCs and their ability to induce pluripotency in human somatic cells (Takahashi et al., 2007; Takahashi and Yamanaka, 2006; Yu et al., 2007), it has been assumed that these genes have important and conserved functions in hESCs. However, due to a great discrepancy in the results reported by different groups (Table 1), the functions of these genes in hESCs remain poorly defined.

Here, by systematically analyzing the phenotypes in knock-down and overexpression cells, we identify three unique features of the pluripotency network in hESCs: (1) NANOG, OCT4, and SOX2 repress primarily embryonic rather than extraembryonic cell fates; (2) the regulatory loop between these genes is uncoupled, allowing the expression of SOX2 in the absence of NANOG and OCT4, as well as expression of NANOG and OCT4 in the absence of SOX2; and (3) instead of working as panrepressors of differentiation, each factor controls specific cell fates. Our studies delineate unique regulatory circuitry that maintains self-renewal and controls lineage commitment in hESCs.

RESULTS

Cell-Line-Specific Usage of Pluripotency Factors in hESCs

To assess the contributions of pluripotency factors to hESC self-renewal, we inactivated each gene using a lentiviral short hairpin RNA (shRNA) vector (Figure 1A). Prior to these experiments, we confirmed that vector-transduced hESCs can be maintained in culture for more than 20 passages, while retaining stable mCherry expression and the ability to form teratomas (data not shown). To ensure efficient gene downregulation, twenty shRNAs were tested for each gene and three constructs with 5- to 10-fold reductions in RNA and protein levels were selected.

Table 1. Summary of Previous Work on NANOG, OCT4, and SOX2 in hESCs

Study	Genes Analyzed	hESC Lines	Culture Conditions	KD Method	Time of Analysis	Markers Observed
(Babaie et al., 2007)	OCT4	H1	feeder-free; MEF-CM + KSR + bFGF	siRNA transfection	3 days	TE
(Hay et al., 2004)	OCT4	H1, H9	feeder-free; MEF-CM + KSR + bFGF	siRNA transfection	2 days	pEN in H9, but not in H1
(Matin et al., 2004)	OCT4	H7, H14	MEF monolayer KSR + bFGF	siRNA transfection	3 days	TE
(Vallier et al., 2009)	NANOG	H9, hSF6	feeder-free; chemically defined + Activin A + bFGF	stable shRNA	multiple time points	NE
(Zaehres et al., 2005)	OCT4, NANOG	H9	MEF monolayer + KSR + bFGF	lentiviral shRNA	3 and 7 days	for OCT4: none; for NANOG: TE, pEN
(Hyslop et al., 2005)	NANOG	H1, NCL1	MEF monolayer+ KSR + bFGF	siRNA transfection	4 days	TE, pEN
(Fong et al., 2008)	NANOG, OCT4, SOX2	H1, H9	MEF monolayer + KSR + bFGF	siRNA transfection	4 days	TE, pEN for all three factors
(Adachi et al., 2010)	SOX2	khES1, HES3	MEF monolayer + KSR + bFGF	siRNA transfection	3 days	TE, EN

TE, trophectoderm; pEN, primitive endoderm; NE, neuroectoderm; MEF, mouse embryonic fibroblast; MEF-CM, MEF-conditioned media; KSR, KnockOut Serum Replacement (Sigma).

Functional and molecular differences among hESC lines have been documented (Bock et al., 2011). We considered that the requirements for pluripotency factors may also vary. To determine which hESC lines would be the most informative, we compared gene expression profiles of 20 hESC lines, including H1, H7, H9, and 17 HUES hESC lines (Bock et al., 2011). Principal component analysis (PCA) revealed that the majority of HUES lines clustered with H7 cells, whereas H1 and H9 cells were distant from each other and from H7 cells, defining three potentially distinct classes of hESCs (Figure 1B). Given these results, we analyzed H1, H7, and H9 cells.

To measure self-renewal in shRNA-transduced (KD) hESCs, we performed a fluorescence-based competition assay (Figure 1C). mCherry-positive (mCherry⁺) KD hESCs were mixed with hESCs transduced with the nontargeting GFP-expressing vector (GFP⁺), and mCherry⁺/GFP⁺ ratios were measured for six passages. As expected, the mCherry⁺/GFP⁺ ratio remained constant in control cocultures. Rapid decreases in mCherry⁺/GFP⁺ ratios were observed for all NANOG and OCT4 shRNAs in all three lines (Figure 1D). However, in SOX2-KD cocultures, the mCherry⁺/GFP⁺ ratio decreased only in H1 cells. That NANOG and OCT4 are required for self-renewal contrasts with SOX2, which may be dispensable in certain hESC lines.

To determine whether the loss of pluripotency factors induces differentiation, we analyzed pluripotency profiles (Figures 1E–1G). Eight days after transduction, NANOG-KD H1 cells downregulated the cell surface antigen TRA1-81 and OCT4; however, the retention of SOX2 (Figure 1E) suggested a conversion into the neuroectoderm (NE). In contrast, SOX2-KD cultures maintained unaltered marker levels, suggesting that these cells remain undifferentiated (Figure 1F). Similar results were obtained for NANOG and SOX2 in all three cell lines. In contrast, the marker profiles in OCT4-KD cells were cell line dependent (Figure 1G). OCT4-KD H1, H7, and H9 cells each downregulated SSEA-4, TRA1-81, and NANOG; however, H9 cells retained

high levels of SOX2, suggesting that, similar to NANOG-KD cells, OCT4-KD H9 cells convert into NE.

To further investigate lineage induction, factor-specific differentiation signatures were identified by microarrays. The expression of 1,952 genes was perturbed more than 2-fold in at least one experiment (Table S1 available online); this data set was visualized by PCA (Figure 1H). As expected, SOX2-KD H1 cells clustered with control cells, while significant changes were observed in NANOG-KD and OCT4-KD H1 cells. Interestingly, NANOG-KD and OCT4-KD cells diverge on the PCA plot, suggesting the emergence of different cell types. OCT4-KD H9 cells were also profiled and clustered with NANOG-KD H1 cells and away from OCT4-KD H1 cells. Taken together, these results suggest that the usage of pluripotency factors in hESCs differs from mESCs and varies among hESC lines.

Cell-Line-Specific Phenotypes in OCT4-KD hESCs Are Due to Differential Activation of the BMP4 Pathway

To determine which differentiation programs are activated in OCT4-KD hESCs, we analyzed lineage markers in these cells. OCT4 KD in H1 cells resulted in a complex marker pattern. Extra-embryonic ectoderm (exEC) markers CDX2, HAND1, GATA3, TFAP2A, and KRT7 were upregulated. Markers expressed in both extraembryonic and embryonic ectoderm (eEC), such as DLX3, DLX5, and MSX1/2, were also induced (Figures 2A and S1, available online). pEN marker SOX7, GATA6, and GATA4 were also induced, whereas primitive streak (PS) and NE markers were not. A similar pattern was observed in OCT4-KD H7 cells (Figures 2A, S1). Downregulation of OCT4 in H9 cells, however, did not induce markers observed in H1 and H7 cells, but rather the NE marker PAX6. In the presence of serum, another NE marker, ZIC1, and the neural rosette marker PLZF were also induced (Figures 2A, S1). Therefore, OCT4 is required to repress both extraembryonic and epiblast-derived lineages, and the exact differentiation programs activated by OCT4

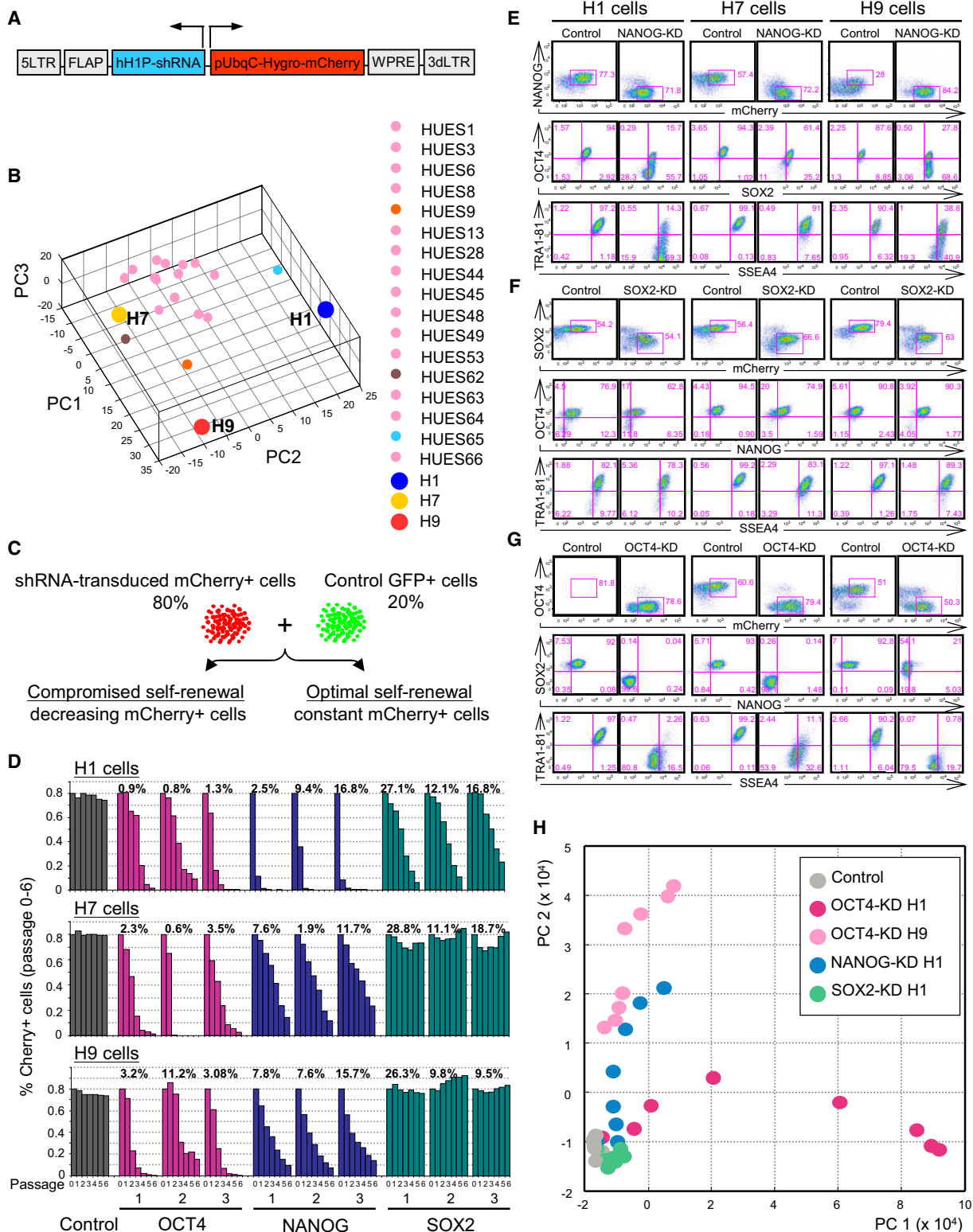


Figure 1. Requirements for NANOG, OCT4, and SOX2 in hESC Self-Renewal

(A) Lentiviral vectors expressing both shRNA and Hygromycin-mCherry fusion protein were utilized in this study.

(B) PCA of gene expression in 20 hESC lines from Bock et al. (2011) identified three hESC classes. Three PCs that account for 52% of the variance in the data set are shown.

(C) A fluorescence-based competition assay was designed to measure changes in self-renewal.

downregulation are specified by additional hESC-line-specific regulatory components.

To identify such cell-line-specific components, we compared the microarray signatures of OCT4-KD H1 and H9 cells. We found that BMP4, which induces exEC markers in hESCs (Xu et al., 2002) and its downstream targets—ID1, ID2, and SMAD6—were upregulated immediately after OCT4 KD in H1 cells, but not in H9 cells (Table S1, Figure 2A). To determine whether BMP4 signaling is required for the induction of markers observed in OCT4-KD H1 cells, cells were treated with the BMP4 inhibitor NOGGIN. Low concentrations of NOGGIN permitted induction of some exEC markers, but not pEN markers, whereas high concentrations of NOGGIN led to a complete switch toward the NE. Conversely, an addition of BMP4 to the culture media was sufficient to block NE induction in H9 cells and to induce markers observed in H1 and H7 cells (Figure 2A).

Taken together, these findings suggest that BMP4 is a critical cell-line-specific target of OCT4 in hESCs that drives the induction of exEC and pEN differentiation programs in OCT4-KD cells. In the absence of BMP4 signaling, NE differentiation is activated as a default lineage choice.

OCT4 Expression and BMP4 Signaling Combinatorially Specify Cell Fates in hESCs

To determine whether BMP4 induction alone accounts for the marker pattern observed in OCT4-KD H1 cells, we compared microarray profiles of OCT4-KD and BMP4-treated H1 cells. The 1,395 OCT4-responsive and 1,466 BMP4-responsive genes were identified and showed significant overlap (Figure 2B, Table S1). The overlapping gene set was enriched for gene ontology (GO) functions associated with differentiation and development, and contained many markers that are expressed in both extraembryonic and PS lineages, including CDX2, HAND1, TFAP2A, and GATA3. OCT4-responsive genes that did not respond to BMP4 treatment were enriched for GO functions associated with cell proliferation and metabolism and contained exEC markers DLX3 and KRT7. In contrast, BMP4-responsive genes that were not OCT4 responsive were enriched for functions associated with development and differentiation, and contained the key ME-specifying genes T, MIXL1, and EOMES. Thus, gene expression data suggested that in hESCs, BMP4 signaling directs ME differentiation when combined with high levels of OCT4, but results in exEC/pEN differentiation when combined with low levels of OCT4.

To further test this hypothesis, we monitored marker induction in BMP4-treated and OCT4-KD H1 cells by immunofluorescence staining. With BMP4 treatment, both CDX2 and T were induced at day 2 when OCT4 levels remained high (Figure 2C). Costaining with T, CDX2, and OCT4 antibodies revealed that all T⁺ cells were also positive for CDX2 and OCT4 (Figures 2G and 2E).

No T⁺ cells were found in OCT4-KD H1 cultures (Figure 2D). By day 4, BMP4-treated cells downregulated OCT4 and T and upregulated HAND1, which was largely expressed in CDX2^{Low} cells (Figures 2C, 2H, and 2E). HAND1 functions in exEC, but it is also a marker of cardiac and lateral ME (Barnes et al., 2010), which lacks CDX2. Therefore, the HAND1⁺ subset in BMP4-treated cultures likely represents ME lineage. Thus, BMP4 signaling combined with high OCT4 levels drives ME differentiation, which agrees with the recent report showing that BMP4-treated hESCs adopt ME rather than TE fates (Bernardo et al., 2011).

The lack of T, MIXL1, and EOMES expression in OCT4-KD cells argues against the possibility that these cells are ME derived. CDX2+HAND1⁺ cells, a phenotype unique for the exEC in the mouse, were the most abundant cell population in these cultures (Figures 2D, 2H, and 2E) and were accompanied by clusters of KRT7⁺ cells that resembled multinucleated syncytiotrophoblast (Figure 2F, top). Although a small number of KRT7⁺ cells could be obtained in BMP4-treated cultures in the presence of serum, these cells resemble mesenchymal rather than TE-like cells (Figure 3F, bottom). However, while OCT4-KD cultures expressed many exEC markers, EOMES and ELF5, two genes that are required for the long-term proliferation of trophoblast stem cells (TSCs) in the placenta (Hemberger et al., 2010), were not induced. Furthermore, the proximal promoter of the *ELF5* gene remained hypermethylated in a large fraction of OCT4-KD H1 cells (Figure S2A). Therefore, although loss of OCT4 in H1 cells generates a phenotype resembling exEC, it is insufficient for establishing TSCs from hESCs.

In addition to HAND1⁺ cells that accounted for nearly 90% of OCT4-KD H1 culture, about 8% of day 4 OCT4-KD H1 cultures also expressed pEN marker SOX7 (Séguin et al., 2008), which was not detected in BMP4-treated cells (Figures 2C, 2D, 2E, and 2I). Given that exEC and pEN do not share a common precursor, we questioned whether the observed marker pattern reflects heterogeneity within the H1 culture, in which case, the exEC and pEN genes would be induced in nonoverlapping subsets of cells. Immunofluorescence staining with SOX7 and CDX2 antibodies revealed that many SOX7 cells coexpressed CDX2 (Figures 2I and 2E). Similarly, GATA6⁺ cells weakly expressed exEC marker TFAP2A at day 6 (Figure S2B). Thus, in hESCs, OCT4 represses both exEC and pEN genes in the same cell.

In OCT4-KD H9 cells, PAX6+SOX2⁺ cells were the main cell population, confirming the result of qPCR analyses (Figure 2J).

Given that ME differentiation is not observed in OCT4-KD hESCs, we asked whether high levels of OCT4 are also required for the differentiation into definitive endoderm (dEN). Inactivation of OCT4 significantly reduced the number of SOX17⁺ dEN progenitors compared to control cells (Figure S4A).

(D) The proportions of KD cells in competition cocultures were measured at each passage in H1, H7, and H9 cells. Three shRNA constructs were assayed for each gene. RNA levels of targeted genes were quantified by qPCR and are shown as a percentage of wild-type levels above each graph.

(E–G) Protein levels of pluripotency markers in NANOG (E), SOX2 (F), and OCT4-KD (G) cells were assayed 8 days after shRNA transduction. Cells were costained with OCT4, NANOG, SOX2, SSEA-4, and TRA1-81 antibodies and analyzed by FACS. The transduced cells were gated based on mCherry expression (top row). The middle and bottom rows show expression of pluripotency markers in the gated subsets.

(H) Factor-specific differentiation signatures were identified by microarrays. RNA from KD and control cells was collected daily during an 8 day time course and used to interrogate Illumina arrays. 1,952 nonredundant genes with expression perturbed more than 2-fold in at least one experiment were used in PCA analysis. Two PCs that account for 58% variability in this data set are shown.

See also Table S1.

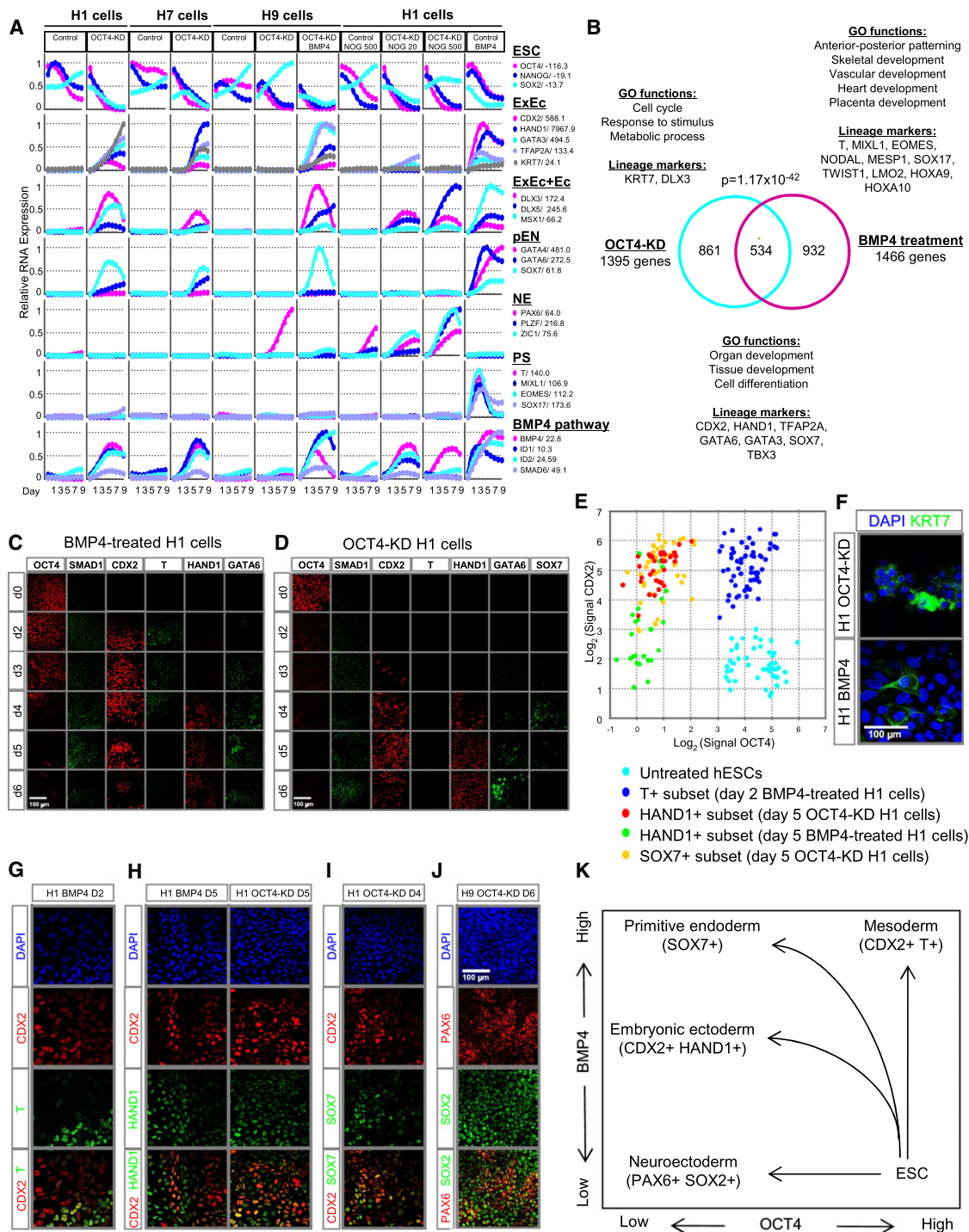


Figure 2. OCT4 Regulates, and Interacts with, the BMP4 Pathway to Specify Distinct Developmental Programs

(A) Cell-line-specific differentiation signatures were observed in OCT4-KD hESCs. Expression of lineage markers and BMP4 pathway components was analyzed in H1, H7, and H9 cells. After shRNA transduction, cells were maintained in mTeSR media with or without cytokines. RNA samples were collected as a 9 day time course and analyzed by qPCR. Data was normalized to GAPDH and scaled to the maximum value of 1. Marker induction levels calculated as ratios between the maximum and minimum expression are shown next to marker names.

(B) OCT4-KD and BMP4-treated H1 cells exhibit overlapping gene expression signatures. RNA samples for each treatment were collected as an 8 day time course and used to interrogate Illumina arrays. Genes upregulated or downregulated at least 2-fold between BMP4-treated and control time courses were defined as BMP4 responsive. OCT4-KD data set contain genes differentially expressed between OCT4-KD and control time courses.

(C and D) Marker profiles of BMP4-treated (C) and OCT4-KD (D) hESCs suggest induction of different lineages. Cells were maintained in mTeSR media and analyzed using the indicated primary antibodies.

Taken together, these findings suggest that OCT4 and BMP4 combinatorially define four cell fates (Figure 2K). High levels of OCT4 permit hESC self-renewal in the absence of BMP4 but promote ME differentiation in the presence of BMP4. Low levels of OCT4 induce NE differentiation in the absence of BMP4 but activate exEC and pEN genes in the presence of BMP4.

Low Levels of SOX2 Are Permissive for hESC Self-Renewal

SOX2-KD hESCs exhibit an unaltered pluripotency profile (Figure 1). To test the long-term self-renewal of SOX2-KD hESCs, single-cell-derived clones with very low SOX2 levels were established (Figure 3A) and evaluated by morphology and SSEA-4 expression. Although there was an increase in differentiated SSEA-4⁺ cells in SOX2-KD clones, distinct SSEA-4⁺ populations were maintained for more than 20 passages (Figure 3B). Microarray profiling of SSEA-4⁺ cells FACS-purified from SOX2-KD clones identified no changes in the expression of pluripotency markers NANOG and OCT4. However, an increase in the expression of T, FOXA2, and EOMES was detected, suggesting that hESCs with low SOX2 levels may be biased toward the PS (Figure 3C). Indeed, under the self-renewing conditions, ~8% of cells in SOX2-KD cultures express either T or FOXA2 (Figure S3A). To investigate this lineage bias further, SOX2-KD cells were differentiated into PS derivatives. In the presence of ACTIVIN A, which promotes dEN differentiation (Borowiak et al., 2009), both control and SOX2-KD clones gave rise to SOX17⁺ cell clusters (Figure 3D). However, SOX2-KD clones were highly efficient in generating SOX17⁺ clusters in the absence of ACTIVIN A, unlike the control clones. To test ME potential of SOX2-KD clones, cardiomyocyte differentiation was induced in monolayer cultures and scored by the expression of cardiomyocyte marker cTNT (Osafune et al., 2008). No cTNT⁺ cells were detected in the control cultures, while SOX2-KD clones gave rise to multiple cTNT⁺ clusters (Figure 3E). Hematopoietic differentiation was induced by sequential treatments with ACTIVIN A, BMP4, and VEGF, and scored by the expression of CD34 and CD31 (Lu et al., 2008). A 5- to 10-fold increase in CD34⁺ as well as CD34⁺CD31⁺ cells was observed in SOX2-KD cultures relative to the control (Figures 3F and 3G).

Elevated PS markers and their enhanced ability to form PS derivatives in functional assays raised a question of whether SOX2-KD cells are operationally pluripotent or instead represent self-renewing PS progenitors incapable of eEC differentiation. This was examined by differentiation into neural and epidermal progenitors in vitro. Although SOX2-KD clones did not form PAX6⁺ cells (Figure S4B), their ability to form TP63⁺ epidermal progenitors was unaffected (Figure 3H). Furthermore, multilineage teratomas consisting of tissue structures derived from three germ layers were obtained for all SOX2 clones tested (Figure 3I).

Taken together, these data suggest that SOX2-KD hESCs maintain pluripotency, albeit less efficiently than control cells.

SOX2 and SOX3 Are Redundant and Repress PS Differentiation

We next tested whether SOX3, which is redundant with SOX2 in the NE (Graham et al., 2003), is expressed in hESCs and could compensate for the loss of SOX2. The expression of SOX3 in hESCs was low (Figures 4A–4C) and its depletion by shRNAs did not affect hESC morphology or the marker profile (Figures 4E, S3B, S3C). However, SOX3 was induced in SOX2-KD cells (Figures 4A–4C). Simultaneous depletion of SOX2 and SOX3 resulted in hESC differentiation, which was most pronounced in H1 cells (Figure 4D). Lineage analysis in SOX2/3-KD H1 cells revealed the induction of dEN markers GATA6, GATA4, FOXA2, and SOX17 (Figure 4F). CDX2 was also induced in the absence of other exEC genes, as occurs in the gut (Gao et al., 2009), and ME markers T, MIXL1, and EOMES were also induced. By immunofluorescence staining, FOXA2⁺ cells were abundant in SOX2/3-KD cultures and were negative for T, but positive for SOX17, indicating the emergence of distinct PS populations (Figure 4G).

These data suggest an inverse correlation between SOX2 and SOX3 levels in hESCs. The SOX2^{High}SOX3^{Low} cells maintain a high rate of self-renewal and the potential to contribute to both NE and PS lineages, whereas SOX2^{Low}SOX3^{High} cells have a reduced capacity to self-renew, but retain pluripotency. Downregulation of SOX3 in these cells initiates commitment to PS (Figure 4H).

NANOG Represses NE and Neural Crest Commitment in hESCs

Two different phenotypes in NANOG-KD hESCs have been reported: with MEF-conditioned media and siRNAs, exEC and pEN markers were induced, while with chemically defined media and stable shRNAs, a NE phenotype was observed (Table S1). FACS profiling of NANOG-KD cells (Figure 1E) identified the OCT4⁺/SOX2⁺ phenotype compatible with NE differentiation. However, significant proportions of NANOG-KD H1, H7, and H9 cultures were OCT4^{Intermediate}/SOX2⁺, representing either a stage that precedes NE or a non-NE lineage. To determine the marker profile of NANOG-KD cells, we performed qPCR analyses of H1, H7, and H9 cells in MEF-conditioned media, which is permissive for a broad spectrum of lineages. While markers of ExEC, pEN, and PS were undetected, various marker combinations of eEC were induced in all three cell lines (Figure S1). In addition to NE markers PAX6, ZIC1, and PLZF, NANOG-KD hESCs induced neural crest (NC) markers GBX2 and PAX3, suggesting that NANOG represses a broader range of eEC derivatives. H7 cells were positive for OTX2, which, in the mouse, is expressed

(E) Putative precursor subsets in BMP4-treated and OCT4-KD hESCs differ in OCT4 and CDX2 levels. Protein levels in individual cells were quantified using confocal images costained with CDX2, OCT4, DAPI, and a subset-specific marker. Cells with the highest levels of the subset-specific marker were selected and expression values for the other three channels were extracted using ImageJ software. Background-corrected Log₂-transformed signals are displayed as a scatter plot.

(F) Morphologies of KRT7⁺ cells in OCT4-KD (top) and BMP4-treated (bottom) cells suggest exEC versus ME differentiation. OCT4-KD cell were maintained in mTeSR media and BMP4-treated cells were maintained in MEF-conditioned media for 7 days.

(G–J) Costaining identifies putative precursor populations in BMP4-treated and OCT4-KD hESCs.

(K) A model of lineage specification by OCT4 and BMP4.

See also Figures S1, S2, S4, and Table S1.

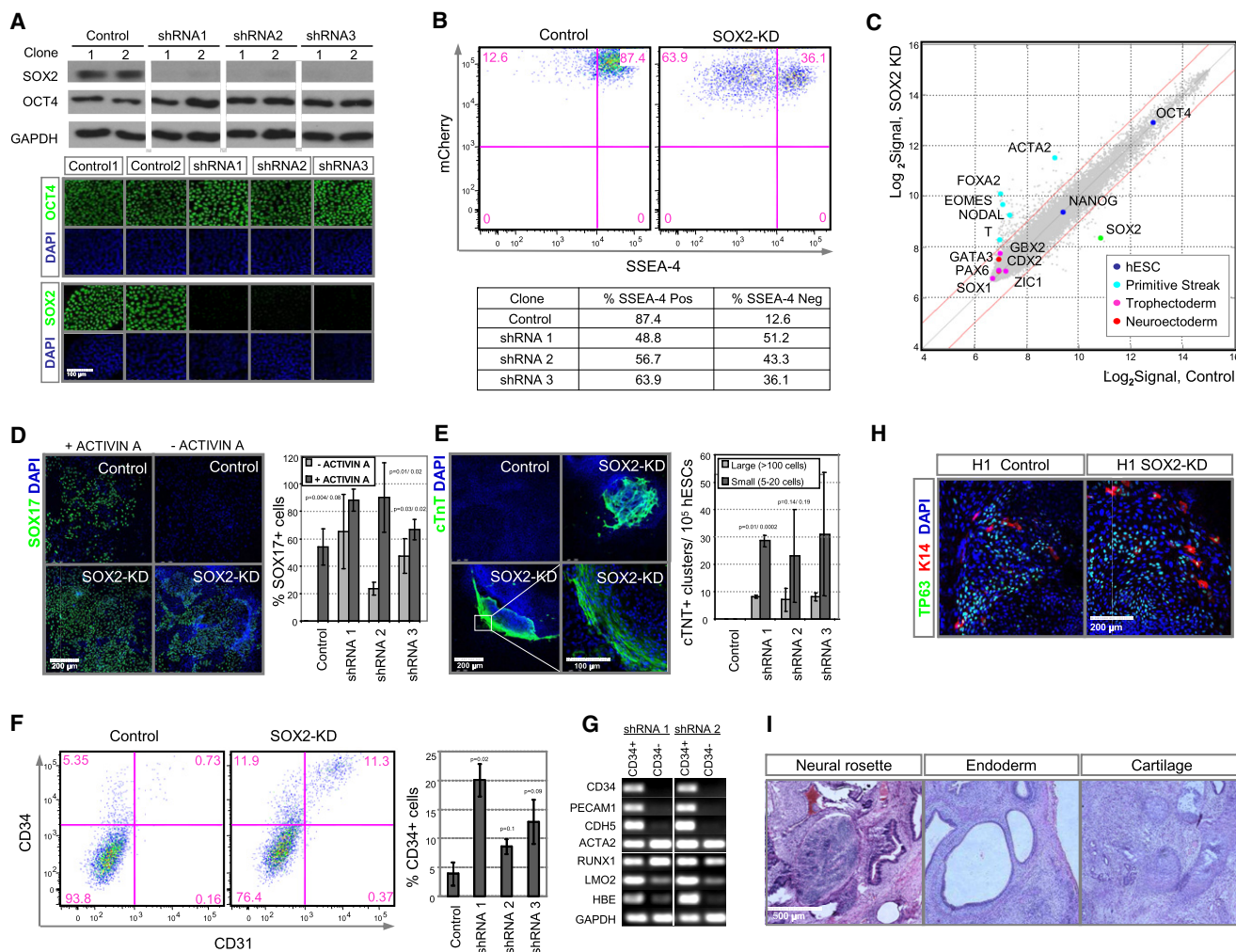


Figure 3. Low SOX2 Levels Are Permissive for Self-Renewal but Facilitate PS Differentiation

(A) Single-cell-derived clones with very low SOX2 levels were generated from SOX2-KD cells. SOX2 and OCT4 levels were assayed by western blotting (top) and immunofluorescence (bottom).

(B) The SSEA-4⁺ cell population is maintained in SOX2-KD clones. FACS plots of SSEA-4 expression and the proportions of SSEA-4⁺ and SSEA-4[−] cells at passage 10 are shown.

(C) SOX2-KD clones maintain expression profiles of undifferentiated hESCs. A scatter plot of expression data from the SOX2-KD microarray experiments is shown. Each point corresponds to the Log₂-transformed expression value of one probe in SOX2-KD and control clones. The black diagonal line represents a 1:1 expression ratio between SOX2-KD and control clones. The red lines demarcate the 2-fold expression interval.

(D–F) Enhanced dEN (D), cardiomyocyte (E), and hematopoietic (F) differentiation is observed in SOX2-KD clones. The numbers of SOX17⁺ cells, cTNT⁺ clusters, and CD34⁺ cells are shown in the bar plots next to the images. The average values and standard errors shown represent data from two replicate experiments.

(G) Both hematopoietic and endothelial markers are induced in the FACS-purified CD34⁺ cell subset obtained from SOX2-KD clones.

(H) SOX2-KD clones form TP63⁺ and KRT14⁺ epidermal progenitors in vitro. Cells were grown in mTeSR media supplemented with BMP4 for 3 days, and then in serum-containing media for 7 days.

(I) ME, dEN, and NE-derived structures are present in teratomas from SOX2-KD clones. Hematoxylin and eosin-stained tissue sections are shown.

See also Figures S1, S3, S4 and Table S1.

in the anterior epiblast prior to PAX6 induction (Kurokawa et al., 2004). H1 cells, on the other hand, were negative for OTX2 but had higher levels of GBX2, which, in addition to its role in NC development (Li et al., 2009), is also a posterior NE marker (Martinez-Barbera et al., 2001) (Figures 5A–5C). Thus, downregulation of NANOG in different hESC lines induces NE subsets with different anterior-posterior identities. To test whether NE and NC can be selectively expanded from the precursors generated in NANOG-KD cultures, H1 cells were briefly treated with different

cytokines. Treatment with NOGGIN enhanced NE signature (Figures 5D and 5E), whereas bFGF-treated NANOG-KD cells resembled NC cells (Figures 5D and 5G). Stimulation with BMP4 induced exEC and pEN lineage markers in both NANOG-KD and control cells (Figures 5D and 5H), suggesting that BMP4 can override early stages of NE/NC commitment in NANOG-KD cells, similar to its effect on OCT4-KD H9 cells.

To determine whether low NANOG levels alter the potential of hESCs to contribute to PS, NANOG-KD H1 cells were tested in

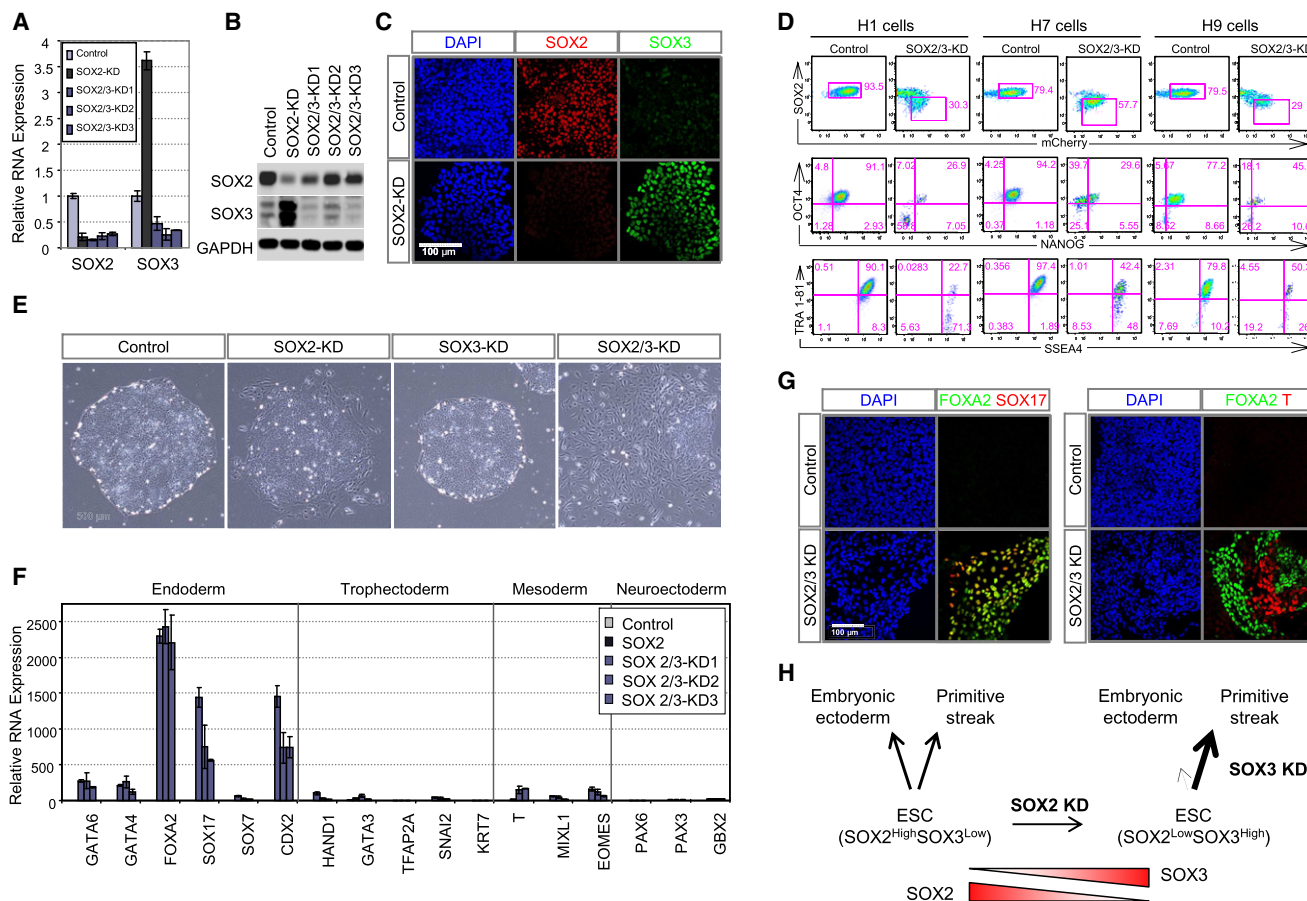


Figure 4. SOX3 Supports hESC Self-Renewal in the Absence of SOX2

(A–C) SOX3 is induced in SOX2-KD hESCs. RNA levels of pluripotency markers (A) were quantified by qPCR and are shown as fold change relative to the control. SOX2 and SOX3 protein levels were assayed by western blotting (B). Induction of SOX3 in SOX2-KD clones was confirmed by immunofluorescence staining (C). Prior to the assays, cells were maintained in mTeSR media for 6 days. Average values and standard errors from two replicate experiments are shown.

(D) Simultaneous depletion of SOX2 and SOX3 induces hESC differentiation. Pluripotency markers were assayed by FACS. The transduced cells in control (left) and KD (right) cultures were gated based on mCherry expression (top). The middle and bottom rows represent marker expression in the gated subsets.

(E–G) SOX2/3-KD cells differentiate into ME and dEN. Cell morphologies (E) and RNA expression of lineage markers (F) were examined. Expression is shown as fold change relative to the control. Average values and standard errors represent two replicate transduction experiments. Distinct PS populations are present in KD hESCs (G). Cells were replated at low density 2 days after shRNA transduction and grown in mTeSR media for 6 days.

(H) SOX2 and SOX3 levels define a putative hESC subset with a bias toward the PS.

See also Figures S3 and S4.

directed differentiation assays. Neither dEN nor hematopoietic differentiation was affected (Figures S4A, S4C). Overall, these data suggest that in hESCs, NANOG is a specific repressor of NE/NC cell fates.

Overexpression of NANOG, OCT4, and SOX2 Does Not Induce hESC Differentiation

In mESCs, a number of pluripotency regulators, including Oct4 and Sox2, function as lineage specifying factors (Loh and Lim, 2011). To determine whether NANOG, OCT4, and SOX2 function as lineage specifiers in hESCs, overexpression (OE) studies were performed in H1 cells using lentiviral vectors (Figure 6A). Upregulation of RNA and protein was confirmed for all three genes (Figures 6B, 6C, S5A). In agreement with the previous report (Teo et al., 2011), NANOG-OE cells upregulated dEN markers EOMES, FOXA2, and SOX17; however, only a few

FOXA2+SOX17+ cells were identified by immunofluorescence (Figure S5B). OCT4 and SOX2-OE cells did not induce any lineage markers (Figure 6D). We used microarray profiling to assess the effect of OE on the entire transcriptome (Figure 6E). Only very small sets of genes were perturbed in OE cells. GO analyses identified enrichment for metabolic functions in NANOG and SOX2-OE data sets. Genes broadly associated with cell proliferation and development were upregulated in SOX2-OE cells (Table S1). However, no genes specifically associated with germ layer induction were found in these data sets.

To test whether elevated levels of NANOG, OCT4, and SOX2 alter the ability of hESCs to contribute to specific lineages, we differentiated OE cells into NE and PS. No changes in hematopoietic or cardiomyocyte differentiation were observed (data not shown). However, OE of SOX2 reduced the number of

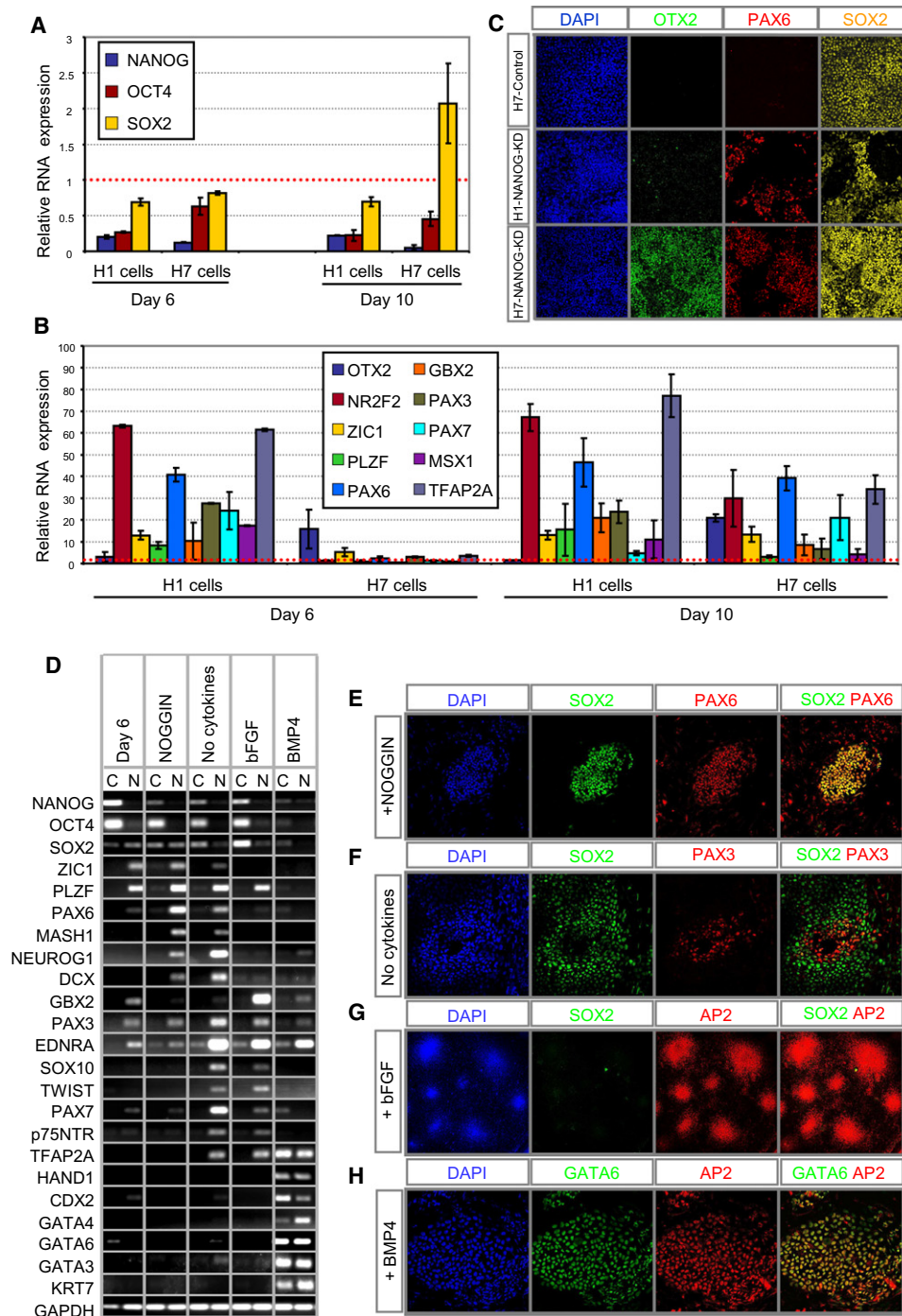


Figure 5. NANOG Represses NE and NC Commitment in hESCs

(A and B) Expression of pluripotency (A) and differentiation (B) markers in KD cells was assessed by qPCR and is shown as a fold change relative to the control (red dotted line). Average values and standard errors are from two replicate experiments.

(C) OTX2 and PAX6 levels were examined in H1 and H7 NANOG-KD cells 12 days after transduction. For (A)–(C) cells were maintained in mTeSR media.

(D) RT-PCR analyses of cytokine-treated cells. NANOG-KD (N) and control (C) H1 cells were maintained in mTeSR media for 6 days, split, and grown in basic neural progenitor cell media supplemented with NOGGIN, bFGF, or BMP4 for 4 additional days.

(E–H) Expression of NE and NC markers was examined by immunofluorescence.

See also Figures S1 and S4 and Table S1.

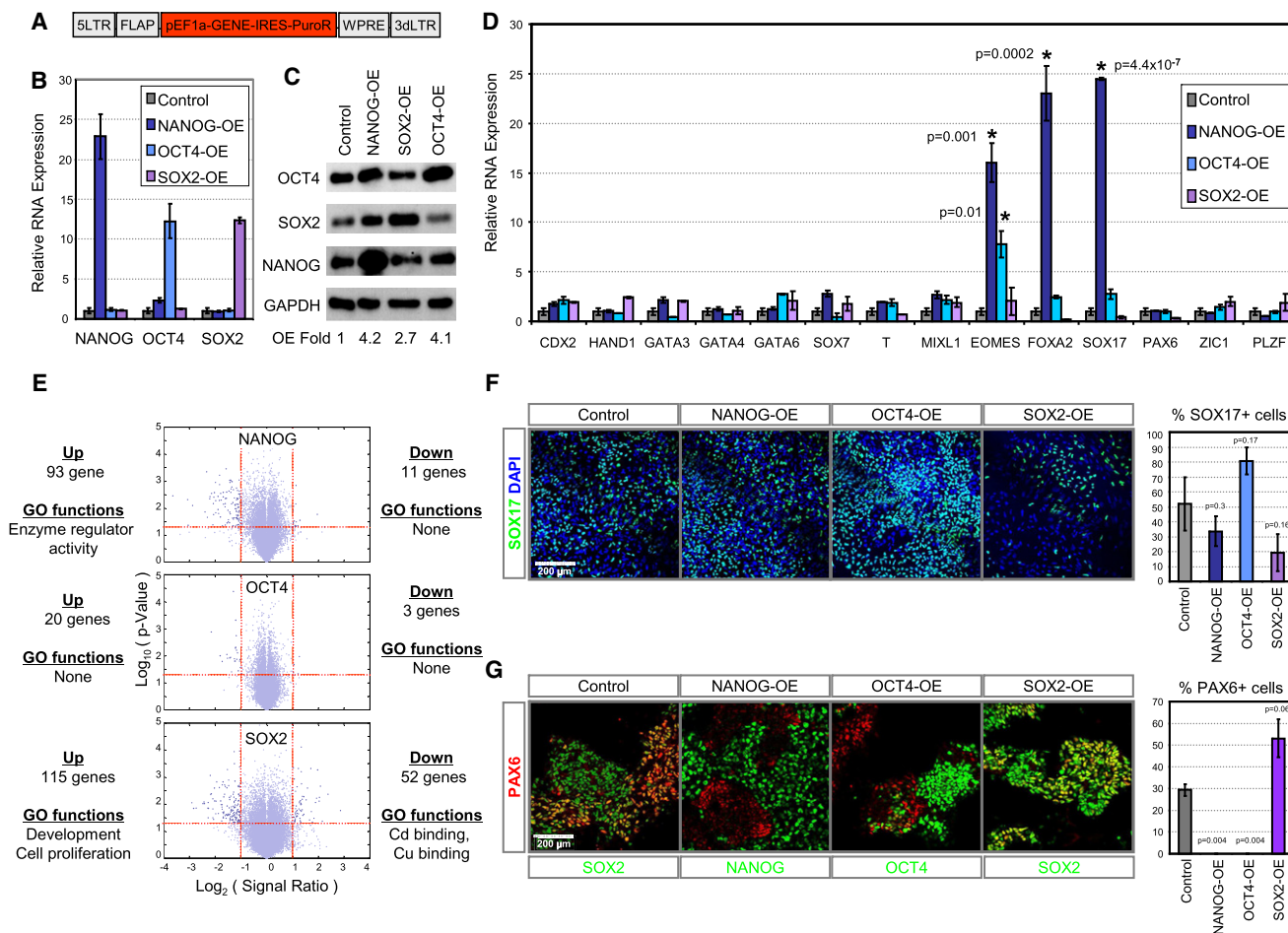


Figure 6. Overexpression of Pluripotency Factors in hESCs Does Not Induce Differentiation

(A) Lentiviral vectors expressing both the gene of interest and the Puromycin-resistance gene were utilized in OE studies.

(B and C) RNA and protein levels were upregulated for all three factors. RNA levels were quantified by qPCR (B), and protein levels, by western blotting (C). Signal intensities are shown as the fold change relative to the control.

(D) RNA expression of germ layer markers was measured by qPCR and is shown relative to the control. Average values and standard errors from two replicate transduction experiments are shown.

(E) OE cells maintain expression profiles of undifferentiated hESCs. A “volcano” plot of expression data for each OE construct is shown. Each point corresponds to one probe, which is displayed based on its Log_2 -transformed fold change value relative to the control and Log_{10} -transformed change p value calculated for two replicate experiments. The vertical red lines demarcate 2-fold expression interval, and the horizontal red line corresponds to a 0.05 false-positive detection rate. The total numbers of upregulated and downregulated genes, and GO groups with statistically significant enrichment, are shown. Microarray data can be found in Table S1. For analyses in (A)–(E), cells were grown in Puromycin-supplemented mTeSR media for 5 days. qPCR data are shown as a fold change relative to the control. Average values and standard errors are calculated from two replicate transduction experiments.

(F) dEN differentiation is enhanced in OCT4-OE hESCs and is inhibited in SOX2-OE cells. Monolayer cultures grown in the presence of ACTIVIN A were costained with SOX17 and DAPI. Proportions of SOX17+ cells in OE and control samples are shown in the bar graph.

(G) NE differentiation is blocked in NANOG and OCT4-OE cells and is enhanced in SOX2-OE cells. Monolayer cultures grown in the presence of GSK-3 β , TGF- β , and γ -secretase inhibitors were costained with PAX6 and factor-specific antibodies. Proportions of PAX6+ cells are shown in the bar graph.

Average values and standard errors shown in (F) and (G) represent data from four replicate experiments. For each experiment, nonoverlapping images were quantified to obtain a total of 1,000 cell counts for each OE construct. See also Figures S4 and S5 and Table S1.

SOX17+ dEN progenitors and enhanced NE differentiation. Conversely, OE of OCT4 enhanced dEN and suppressed NE differentiation (Figures 6F and 6G). NANOG OE resulted in a complete block of NE differentiation (Figure 6G). Given the induction of dEN markers in NANOG-OE cells (Figure 6D) and a recent report suggesting a cell-autonomous role for NANOG in dEN specification (Teo et al., 2011), we were surprised to find a decrease in the number of SOX17+ cells in NANOG-OE

cells (Figure 6F). However, in the gastrula-stage mouse embryos, Nanog is expressed in a posterior exEC adjacent to the PS rather than PS itself (Hart et al., 2004; Morkel et al., 2003), suggesting that instead of being a cell-autonomous factor, NANOG may enhance dEN differentiation via a non-cell-autonomous mechanism.

Overall, both KD and OE data strongly suggest that, in hESCs, NANOG, OCT4, and SOX2 function as differentiation repressors,

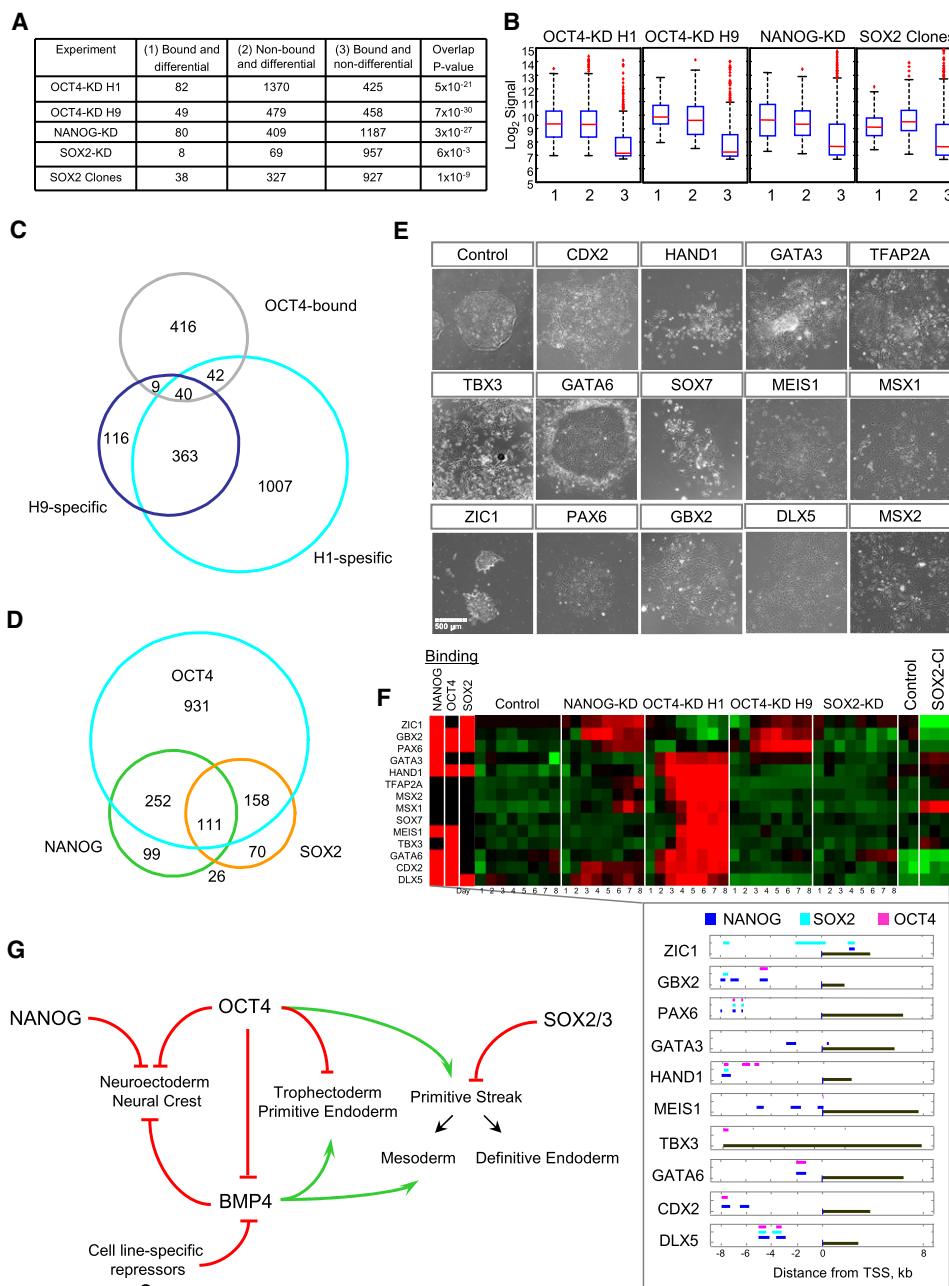


Figure 7. Molecular Networks Regulated by NANOG, OCT4, and SOX2 in hESCs

(A and B) Only a small subset of binding sites is utilized by each factor to directly control gene expression. Binding data from Boyer et al. was compared with sets of genes whose expression was perturbed at least 2-fold in the KD hESCs. Highly significant overlaps between bound and differentially expressed genes were identified in each experiment (A). Box plots of signal value distributions for gene sets defined in (A) are shown for each KD experiment (B). The 75% (blue box) and 95% (black lines) distribution intervals are shown, and the outliers are marked in red. The median distribution values are shown as horizontal red lines.

(C) Lineage commitment in OCT4-KD hESCs is modulated by BMP4 signaling via a coregulation of primary targets. The overlap between the OCT4-KD and OCT4-bound data sets is shown by Venn diagram. Genes were considered differentially expressed in both cell lines if their fold change was greater than 2.0 in at least one of the compared groups and greater than 1.8 in the other groups. These calculations result in increased group sizes compared to the numbers shown in (B). List of coregulated direct targets and their expression profiles in KD time courses are shown in Figure S6.

(D) Gene clusters coregulated by different combinations of pluripotency factors were identified in the microarray data set. Overlaps were calculated as described in (C) and are shown as a Venn diagram. The expression profiles of coregulated targets are shown in Figure S7.

(E) Transcription factors repressed by NANOG, OCT4, and SOX2 induce rapid hESC differentiation when overexpressed individually in H1 cells. Transduced cells were replated at a low density and grown in mTeSR media supplemented with Puromycin for 6 days. No morphology changes were observed in TFAP2C, ISL1, HOXB5, and HESX1 OE cultures.

and that the positive signals that initiate differentiation are mediated via alternative regulatory pathways.

NANOG-OCT4-SOX2-Regulated Molecular Networks in hESCs

To analyze the molecular networks regulated by NANOG, OCT4, and SOX2 in hESCs, we compared our data set of genes differentially expressed during NANOG, OCT4, and SOX2-KD differentiation (Figure 1H) with the data set of genes bound by NANOG, OCT4, and SOX2 (Boyer et al., 2005). To estimate the fraction of all binding sites that are functional in hESCs, we identified the genes bound by each factor and differentially expressed in the knockdown experiments (Figure 7A). More than 80% of all OCT4-bound genes represented on the microarray did not show a measurable change in expression in OCT4-KD hESCs; the majority of these genes were expressed at low levels in both H1 and H9 cells. Similar results were obtained for NANOG and SOX2 (Figures 7A and 7B). Furthermore, for each regulator, the direct targets represented only a minor fraction of all differentially expressed genes. For OCT4, only 5% of genes were direct targets, whereas indirect targets accounted for more than 94% of OCT4-KD differentially expressed genes (Figure 7A). For each of three factors, the direct targets were enriched for GO functions associated with transcriptional control and regulation of differentiation, while the indirect targets were enriched for a broader range of cellular functions, including differentiation and tissue development, signaling, cell division, cell migration, cytoskeleton organization, and regulation of apoptosis (Table S1). Thus, in hESCs, NANOG, OCT4, and SOX2 directly repress or activate small sets of downstream transcriptional regulators, which in turn control a broader range of cellular processes.

Because OCT4 KD induces nonoverlapping phenotypes in H1 and H9 cells (Figure 2), we next identified transcriptional changes that accompany lineage induction in these two cell lines. In at least one cell line, 1,577 genes were differentially expressed (Figure 7C). Among the direct OCT4 targets, 40 genes were coregulated in both H1 and H9 cells, whereas 42 genes were specific in H1 cells and 9 genes were specific for H9 cells (Figures 7C, S6), suggesting that lineage commitment in OCT4-KD hESCs is modulated by external signals, such as BMP4, via a coregulation of primary targets.

In mESCs, Nanog, Oct4, and Sox2 coregulate large cohorts of downstream genes (Ivanova et al., 2006). Surprisingly, a different pattern of target regulation was observed in hESCs. We found that a small group of only 111 genes requires all three factors for the maintenance of proper expression, whereas a larger set of 252 genes is coregulated by NANOG and OCT4, and another set of 158 genes is coregulated by OCT4 and SOX2 (Figures 7D, S7). GO analyses revealed an enrichment of gene-products involved in differentiation and development in all three gene sets. Large regulator-specific gene clusters were also identified. The OCT4 cluster was uniquely enriched for GO functions associated with vascular development, regulation of the cell cycle, and

cell adhesion. The NANOG cluster was enriched for genes associated with neurogenesis and brain development, while the SOX2 cluster was enriched for genes associated with tissue morphogenesis (Table S1).

To define critical differentiation-promoting genes repressed by NANOG, OCT4, and SOX2, we analyzed 18 transcription factors that were induced immediately after the depletion of pluripotency genes (Figure 7E) for their ability to induce hESC differentiation. OE of these genes was performed using lentiviral vectors (Figure 6A). In 14 cases, rapid hESC differentiation was indicated by morphology changes in the cultures (Figure 7E). The majority of these differentiation genes are coregulated by NANOG and OCT4 with an additional input from the BMP4 pathway (Figure 7F).

Taken together, these data suggest a modular pluripotency network in hESCs. Instead of working as a trio, NANOG, OCT4, and SOX2 are recruited into smaller regulatory modules that contribute to the maintenance of self-renewal and the repression of lineage-specific differentiation.

DISCUSSION

In this study, we asked how NANOG, OCT4, and SOX2 maintain hESC pluripotency. Using loss- and gain-of-function assays coupled with transcriptome profiling, we identified requirements for NANOG, OCT4, and SOX2 in hESCs. In mESCs, all three factors are required for self-renewal, and their inactivation results in exEC/pEN differentiation with little, if any, effect on the epiblast-derived cell fates, which are separately repressed by the Esrrb-Tbx3-Tcl1 regulatory module (Ivanova et al., 2006).

We find that the general requirements for NANOG, OCT4, and SOX2 are different between mouse and human ESCs. In hESCs, OCT4 controls both extraembryonic and epiblast-derived cell fates that are chosen in a BMP4-dependent manner. High levels of OCT4 permit hESC self-renewal in the absence of BMP4 and promote PS differentiation in the presence of BMP4. Low OCT4 levels induce NE differentiation in the absence of BMP4 and specify exEC/pEN lineages in the presence of BMP4.

We show that NANOG represses NE/NC differentiation. Furthermore, the outcomes of functional assays suggest that NANOG contributes minimally, if at all, to the repression of other lineages. Thus, instead of being a general repressor of differentiation, NANOG has a lineage-restricted function in hESCs.

In contrast to OCT4 and NANOG, which are required for hESC self-renewal, SOX2 is largely dispensable due to compensation by SOX3. SOX3 expression is low in hESCs, but is upregulated upon SOX2 depletion, permitting hESC self-renewal. SOX2^{Low}SOX3^{High} cells remain pluripotent, but exhibit enhanced spontaneous differentiation toward the PS fates. Simultaneous depletion of both SOX2 and SOX3 results in a loss of self-renewal and ME/dEN differentiation.

Another conclusion from our study is that the usage of pluripotency factors varies significantly among different hESC lines. H1

(F) Microarray profiles of 14 regulators of differentiation in 8 day differentiation time courses. The binding of pluripotency factors is shown in the box below the heat maps. No binding was observed for TFAP2A, MSX1, MSX2, and SOX7. Microarray data for each gene were normalized to the average value 0 and standard deviation 1 and are displayed as heat maps with low expression shown in green and high expression shown in red.

(G) Model of cell fate regulation by NANOG, OCT4, and SOX2/3 in hESCs. See also Figures S6 and S7 and Table S1.

and H7 cells require OCT4 to repress BMP4 expression, whereas an OCT4-independent mechanism operates in H9 cells. H1 cells are more sensitive to SOX2/3 inactivation than H7 and H9 cells. Furthermore, a line-to-line variability in eEC patterns in NANOG-KD hESCs suggests formation of NE subsets with different anterior-posterior identities. Because no such variability has been reported for mESCs, we propose that in contrast to *Lif/Stat3* and *Bmp4/Smad1* pathways that support a unique ICM-specific state, a broader range of developmental states, represented by individual hESC lines, are supported by *FGF2/MAPK* and *TGF β /SMAD2/3* signaling. The ability of H1 and H7 cells to induce BMP4 and other exEC markers in response to OCT4 KD suggests that these hESC lines represent an early developmental state that retains a limited capacity to activate extraembryonic differentiation programs. In the mouse, *Bmp4* expression is detected in both ICM and polar TE and later is excluded from the epiblast (Coucouvanis and Martin, 1999). In contrast, *Oct4* is upregulated to its maximum level in the late epiblast (Han et al., 2010), suggesting that OCT4 may directly repress *Bmp4* during the ICM-to epiblast transition. OCT4 binding at the *BMP4* locus in hESCs (Figures S2C, S2D) supports this model. In contrast to H1 and H7 cells, H9 cells have lost the ability to regulate endogenous BMP4 via OCT4 and have microarray profiles resembling the initial stages of NE differentiation, a phenotype replicated in NANOG-KD hESCs (Figure 1H), suggesting that these cells may represent later epiblast. Furthermore, *SOX2*^{Low}*SOX3*^{High} cells generated following SOX2 KD likely represent the posterior epiblast. During gastrulation, *Sox2* becomes restricted to the anterior half of the embryo while *Sox3* is upregulated throughout the eEC (Wood and Episkopou, 1999). High *Sox3* expression in epiblast cells adjacent to the PS is needed to antagonize the expression of the transcriptional repressor *Snail*, prevent ingression into the PS territory, and ensure the formation of posterior eEC derivatives (Acloque et al., 2011).

Per the prevailing model, NANOG, OCT4, and SOX2 work as a trio to support each others' expression and that of other self-renewal genes, and to repress differentiation genes (Boyer et al., 2005). Although functional and molecular analyses in mESCs agree with that model (Ivanova et al., 2006), our study identifies differences in the organization and function of the pluripotency network in hESCs. While NANOG and OCT4 are coregulated in all three hESC lines (Figures 1E–1G, 2B), SOX2 is excluded from the loop. Indeed, downregulation of either NANOG or OCT4 in the absence of BMP4 signaling does not affect SOX2 expression and both NANOG and OCT4 levels are unchanged in SOX2-KD cells. Furthermore, a small group of only 111 genes requires all three factors for the maintenance of proper expression, whereas larger groups of genes are regulated by NANOG-OCT4 and OCT4-SOX2 pairs, and by individual factors. Thus, instead of working as panrepressors of differentiation, the pluripotency factors individually repress specific cell fates.

In addition to providing new insights into the regulation of pluripotency, our findings are important for lineage commitment studies in hESCs. The lineage of cells in BMP4-treated hESC cultures has been a longstanding paradox in the field, with the most recent report (Bernardo et al., 2011) suggesting that these cells are of ME, rather than TE, origin as initially proposed

(Xu et al., 2002). We show that BMP4 drives differentiation of both ME and TE, but the exact cell type is specified by the levels of OCT4. Low levels of OCT4 mimic the TE phenotype, but are insufficient to specify TSCs. ELF5, a factor required for the TSC maintenance in human placenta (Hemberger et al., 2010), is not induced and its promoter remains hypermethylated in OCT4-KD hESCs (Figure S2A). This contrasts with mESCs, where downregulation of *Oct4* leads to the establishment of functional TSCs that are able to colonize exEC in chimeras (Niwa et al., 2005). There are three possible explanation why this could be the case. First, the similarity between hESCs and the mouse epiblast suggests that hESCs may have passed the developmental window when functional TSCs can be established. Second, the signaling in hESC cultures may be incompatible with TSC differentiation/maintenance. Third, human TSCs may be specified later in development. The latter hypothesis is supported by the recent finding that, in the cattle, ELF5 is not expressed in TE at the blastocyst stage. Furthermore, a cattle ELF5 BAC reporter is not expressed in mouse TE, even though it is expressed in embryonic skin, another domain of ELF5 expression (Pearton et al., 2011).

In mESCs, multiple transcription factors are required for the maintenance of the pluripotent state (Loh and Lim, 2011). Likewise, additional factors are likely to be required in hESCs; several candidate genes have already been identified (Adamo et al., 2011; Chia et al., 2010; Zhang et al., 2011). It will be important to investigate how these novel regulators control pluripotency and how they are integrated with the NANOG-OCT4-SOX2 network described here.

EXPERIMENTAL PROCEDURES

Cell Culture and Differentiation Assays

hESC lines H1, H7, and H9 were obtained from the YSCC hESC Core under the appropriate MTAs and were maintained as originally described (Thomson et al., 1998). BMP4 induction was performed in mTeSR media. dEN, cardiomyocyte, hematopoietic, and NE differentiation was carried out as previously described (Borowiak et al., 2009; Li et al., 2011; Lu et al., 2008; Osafune et al., 2008).

shRNA Design

shRNAs were designed and cloned as described in Ivanova et al. (2006). The list of shRNAs utilized in this study is available in the Supplemental Information.

Quantitative PCR

Total RNA was reverse transcribed using Superscript II and random primers. qPCR was carried out using iQ SYBR Green Supermix (Bio-Rad). Gene-specific primers utilized in this study are available in the Supplemental Information.

Immunoblotting, Immunofluorescence, and Flow Cytometry

Whole-cell protein extracts were prepared using RIPA buffer. Immunofluorescence staining was performed on paraformaldehyde-fixed cells. For the FACS analyses, single-cell suspension was prepared using Accutase. The list of antibodies and dilution ratios are available in the Supplemental Information.

Gene Expression Microarrays, GO, and PCA

Microarrays were conducted on HumanHT-12 v4 Expression BeadChip arrays. Data was normalized using the cubic spline normalization algorithm in the BeadStudio (Illumina). Differentially expressed genes were identified using $\text{Log}_2\text{FC} > 1$ or < -1 and adjusted with $p < 0.05$. Cluster 3.0 and Java Treeview were used for data visualization. GO was performed using DAVID 6.7 (Huang et al., 2009), and PCA, using MatLab R2010b (MathWorks).

ACCESSION NUMBERS

Microarray data are available in the Gene Expression Omnibus (GEO) database (<http://www.ncbi.nlm.nih.gov/geo/>) under the accession number GSE34921.

SUPPLEMENTAL INFORMATION

Supplemental Information for this article includes Figures S1–S7, Table S1, and Supplemental Experimental Procedures and can be found with this article online at [doi:10.1016/j.stem.2012.02.016](https://doi.org/10.1016/j.stem.2012.02.016).

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